Creating a Spring Boot Microservice (Detailed Guide)

Step 1: Go to Spring Initializr

Open https://start.spring.io/ which is the official tool to bootstrap Spring Boot applications.

Step 2: Select Project Details

Project Setup:

Project: Maven - Used for dependency and build management.

Language: Java - Primary language used in Spring Boot.

Spring Boot Version: 4.0.5 - Choose stable version.

Project Metadata:

Group: com.example - Represents your organization or domain.

Artifact: demo - Name of your project (used in JAR name).

Name: demo - Human-readable project name.

Package Name: com.example.demo - Auto-generated and recommended not to change initially.

Packaging: Jar - Preferred for microservices.

Configuration: Properties or YAML - YAML is cleaner for large configs.

Java Version: 17 or 21 recommended (LTS versions).

Step 3: Add Dependencies

Spring Web: Used to build REST APIs.

Spring Boot Actuator: Provides monitoring endpoints like /actuator/health.

Lombok: Reduces boilerplate code like getters/setters.

H2 Database: In-memory database for development and testing.

Step 4: Generate and Download

Click Generate, download the ZIP, extract it, and open in your IDE.

Step 5: Run the Application

```
./mvnw spring-boot:run
```

Alternatively, run the main class with @SpringBootApplication annotation.

Step 6: Verify Application

Open browser and go to:

```
http://localhost:8080

http://localhost:8080/actuator/health
```

Expected Response:

```
{ "status": "UP" }
```

Best Practices

Use Java 17 or 21 (LTS).

Prefer YAML configuration.

Always include Actuator.

Use Jar packaging.

Maintain clean layered architecture: controller, service, repository, model.

Next Steps

Create REST APIs, connect to MySQL/PostgreSQL, and move towards microservices architecture using Eureka and API Gateway.